



University Institute of Engineering
Department of Computer Science & Engineering

Experiment-6

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1. **Aim of the practical:** To study and implement SQL queries using **joins, conditional logic, and data comparison techniques** to solve real-world database problems involving daily fruit sales and hierarchical employee data.

2. **Questions**

(A) A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day.

Difference = Apples Sold – Oranges Sold

Display the **sale date** and the calculated **difference**. The result should be sorted by sale date in ascending order.

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Sales



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Solution:

sale_date date	diff integer
2020-05-01	2
2020-05-02	0
2020-05-03	20
2020-05-04	-1

```
select
    a.sale_date,
    a.sold_num - b.sold_num as diff
from sales a
join sales b
on a.sale_date = b.sale_date
where a.fruit='apples' and b.fruit='oranges';
```

(B) A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where id represents the current node and p_id represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.
- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the **node ID** along with its category and sort the result by node ID.

id	p_id
1	NULL
2	1
3	1
4	2
5	2

Tree

Solution:

id [PK] integer	type text
1	Root
2	Inner
3	Leaf
4	Leaf
5	Leaf

```
SELECT
  id,
  CASE
    WHEN p_id IS NULL THEN 'Root'
    WHEN id IN (SELECT p_id FROM Tree WHERE p_id IS NOT NULL)
      THEN 'Inner'
    ELSE 'Leaf'
  END AS Type
FROM Tree
ORDER BY id;
```

3. Learning Outcomes:

- Learned to write SQL queries using joins and conditional logic.
- Applied SQL techniques to compare and analyze data.
- Learned to classify hierarchical data into Root, Inner, and Leaf nodes.
- Improved practical skills in solving real-world database problems using SQL.